TEACHER GUIDE

Cambridge International AS & A Level
Business (9609)

Production and capacity



4 Operations management (AS Level)
4.1.2 Efficiency, effectiveness, productivity and sustainability; 4.3.1 Capacity utilisation

A revision lesson. Pupils run a car dealership in firms of four and revise production and capacity on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min	45 min	60 min	75 min

What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal stock

Topic tasks

Each firm gets its own figures. The method is the same.

1 Capacity used % — Round 1

Workshop capacity.

Capacity used, as a %?

Prep bays	**3 bays**
Most one bay can do a week	**10 cars**
Cars prepared a week	**24 cars**

ANSWER

80%

3 × 10 = 30; 24 ÷ 30 × 100 = 80%

Why: It has 6 cars of spare capacity a week.

2 Full capacity risks — Round 1

You run at 100% capacity all year.

Which two problems are most likely?

- **A** No spare time for repairs
- **B** Staff stress and more errors
- **C** A higher fixed cost per car
- **D** Spare capacity is wasted
- **E** Staff with too little to do

ANSWER

✓ No spare time for repairs
✓ Staff stress and more errors

Why: Full capacity spreads fixed costs well, but leaves no slack.

3 Fixed cost per car — Round 2

Three workshops: fixed costs a year, and cars prepared.

Order by fixed cost per car, lowest first?

- **A** £30,000 fixed, 300 cars
- **B** £45,000 fixed, 500 cars
- **C** £20,000 fixed, 160 cars

ANSWER

1. £45,000 fixed, 500 cars
2. £30,000 fixed, 300 cars
3. £20,000 fixed, 160 cars

Why: £90, £100, £125 a car: the biggest costs, spread widest, are cheapest.

Topic tasks

4 New productivity

Round 2

More staff, more cars.

New cars prepared per member of staff?

Cars prepared a week	**18 cars**
Staff	**3 staff**
Extra staff	**3 staff**
Extra cars prepared a week	**30 cars**

ANSWER

8 cars

(18 + 30) ÷ (3 + 3) = 8 cars

Why: Productivity rose from 6 to 8 cars per worker.

5 Meet a short peak

Round 3

Demand is above your capacity for two months only.

How should you meet it, and why?

A Subcontract the extra work

B Build a new workshop

ANSWER

✔ Subcontract the extra work

Why: A short peak does not pay for new capacity.

6 Unit cost when full

Round 3

Run the workshop full.

Fall in unit cost at full capacity?

Fixed costs, a month	**£12,810**
Variable cost per car	**£400**
Cars prepared a month	**15 cars**
Most it can prepare a month	**35 cars**

ANSWER

£488

£12,810 ÷ 15 = £854; ÷ 35 = £366; £854 − £366 = £488

Why: The variable cost stays; only the fixed cost per car falls.

Final questions

Each pupil answers 6 final questions on production and capacity. They come from these 10 questions.

1. A service centre can complete 640 jobs a month but completes 512. What is its capacity utilisation?
   ✔ **80%**

2. A parts team produces 1,440 tested units in 240 labour hours. What is labour productivity?
   ✔ **6 units per labour hour**

3. A dealer considers outsourcing parts deliveries. Which comparison best supports an operational decision?
   ✔ **Total cost, reliability, lead-time risk and capability kept or lost**

4. A bodyshop's practical monthly capacity is 900 repair-hours and booked work uses 765 hours. What is spare capacity as a percentage of practical capacity?
   ✔ **15%**

5. A workshop raises utilisation from 82% to 98%, while waiting time and rework rise sharply. What is the best interpretation?
   ✔ **It is near its limit, so delays rise**

6. Technician productivity differs widely between branches. Which study is most likely to support a fair operational decision?
   ✔ **Compare job mix and hours first**

7. Before a wellbeing redesign a workshop produced 450 standard hours with 90 paid hours; after, 520 with 104. What happened to measured labour productivity?
   ✔ **It remained 5 standard hours per paid hour**

8. A fitting centre's 27 fitters fit 918 tow bars a month. What is labour productivity?
   ✔ **34 tow bars per fitter**

9. A bodyshop can do 250 repairs a month and did 160. What is its capacity utilisation, and the effect on unit costs?
   ✔ **64%: each repair carries more fixed cost than at full capacity**

10. A workshop can service 400 vehicles a month and serviced 300 last month: 75%. Which measure is that?
    ✔ **Its capacity utilisation**